



Inspiring Excellence

Network Layer: IPv4 Functions

Lecture 9 | CSE421 – Computer Networks

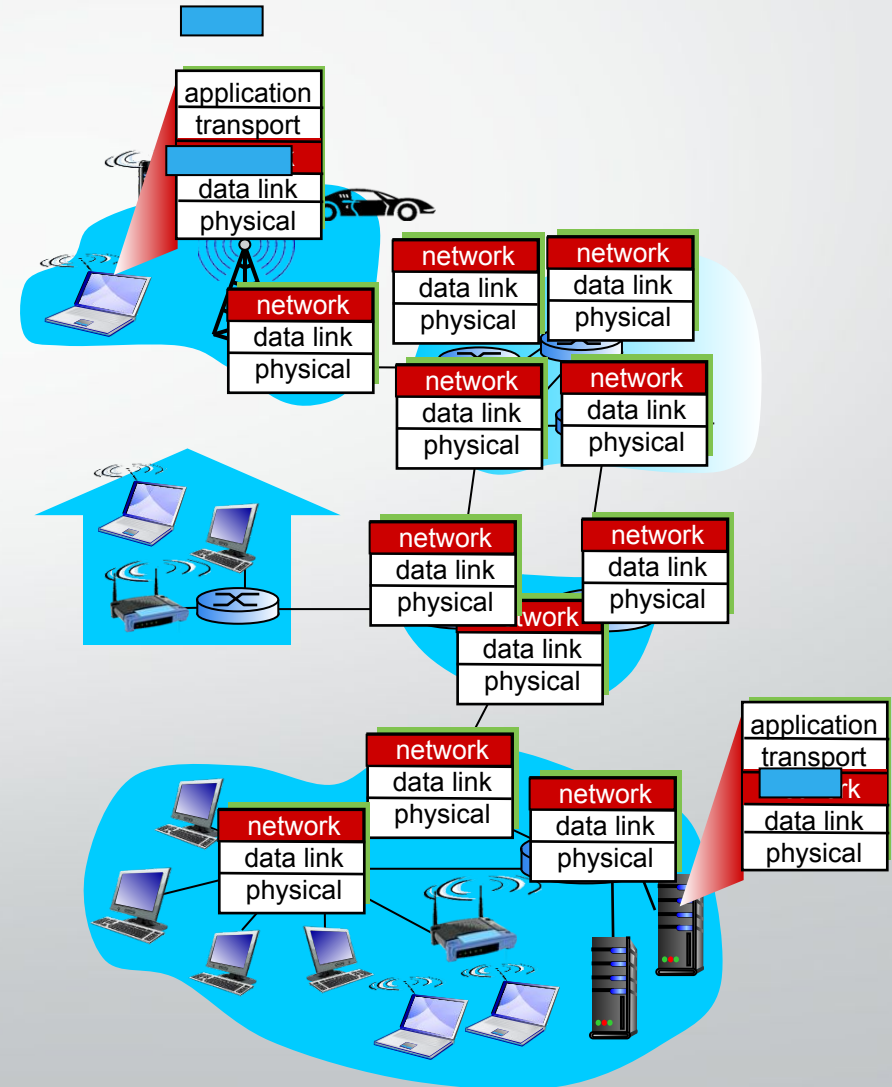
Department of Computer Science and Engineering
School of Data & Science

Objectives

- Short overview of the Network Layer
- Packet Switching: Virtual Circuits & Datagram Networks
- IPv4 Packet Format
- IP Fragmentation & Reassembly
- ICMP
 - Ping
 - Traceroute

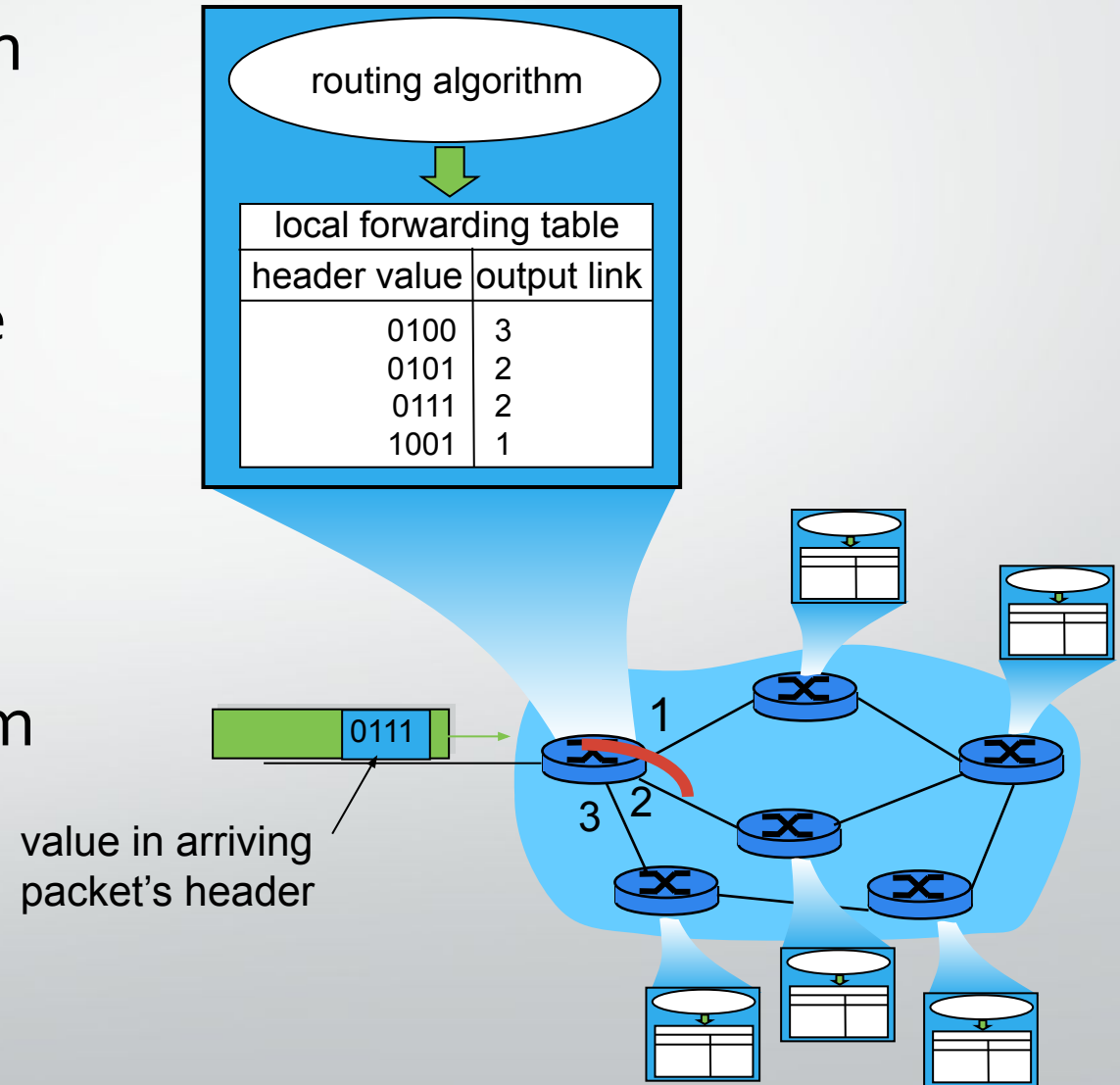
The Network Layer

- Encapsulates data into **packets** on the sending side.
- Network Layer protocols operates on **hosts** and **routers**.
- **Routers** inspect IP header fields for forwarding.
- Delivers segments to the **transport layer** on the receiving side.



Functions of Network Layer

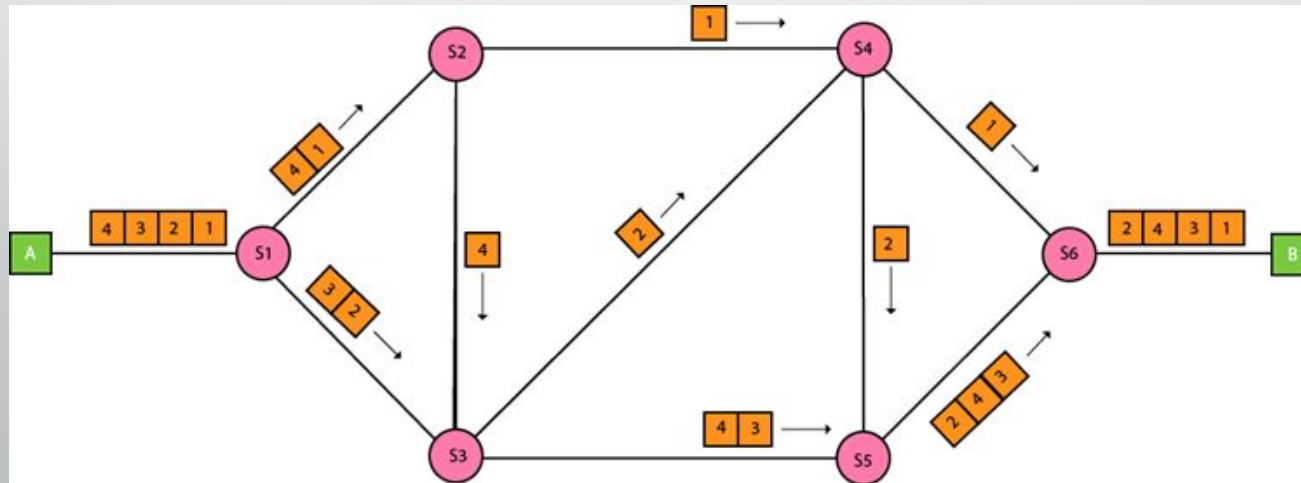
- **Routing:** determine route taken by packets from source to destination
 - The algorithms that calculate the paths are referred to as routing algorithms.
 - Analogy: process of planning trip from source to destination
- **Forwarding:** move packets from router's input to appropriate router output
 - Analogy: process of getting through a single interchange



Packet Switching

Packet Switching

- Part of Network layer
- Packet Switching is a method of transferring data across a network by breaking it into smaller packets.
- Two type of networks based on packet switching
 - **Datagram Networks**
 - **Virtual Circuit Networks**



Datagram networks

- **No Call Setup:**

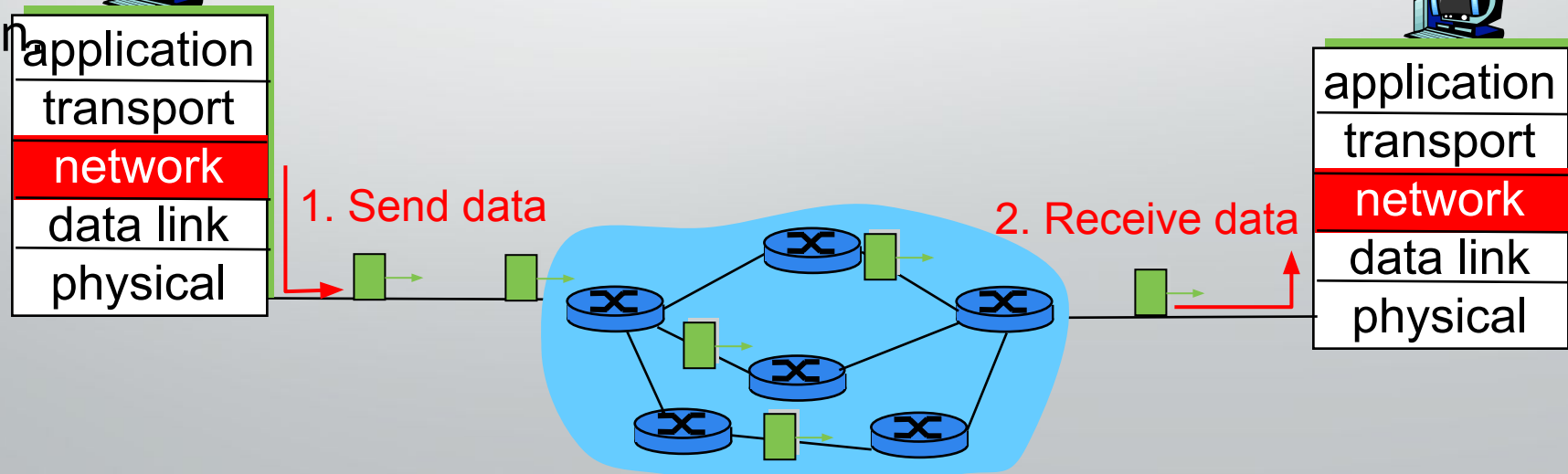
Devices can send data immediately without establishing a connection.

- **Stateless Routers:**

Routers forward packets independently based on their destination IP address.

- **Packet Forwarding:**

Packets from the same source may take different paths to reach the destination.



Virtual Circuits: Signaling Protocols

- **Call Setup:**

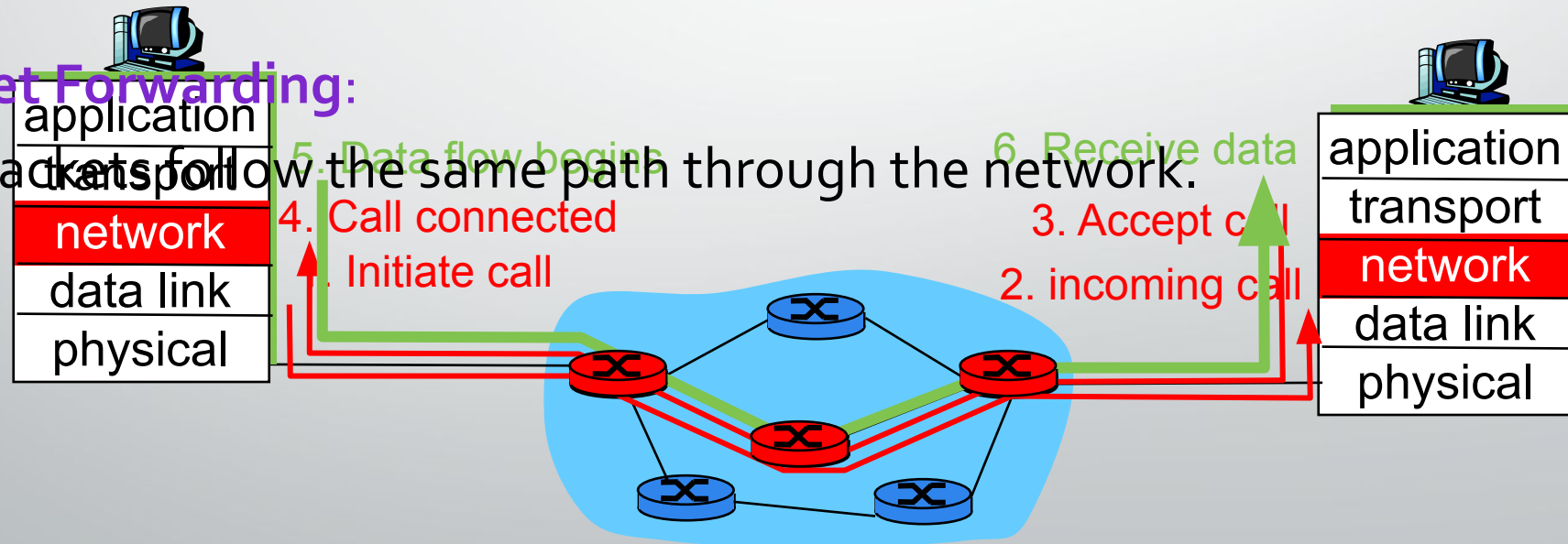
A connection (virtual circuit) is established between sender and receiver before data transfer.

- **Stateful Routers:**

Routers maintain information about active connections (virtual circuits).

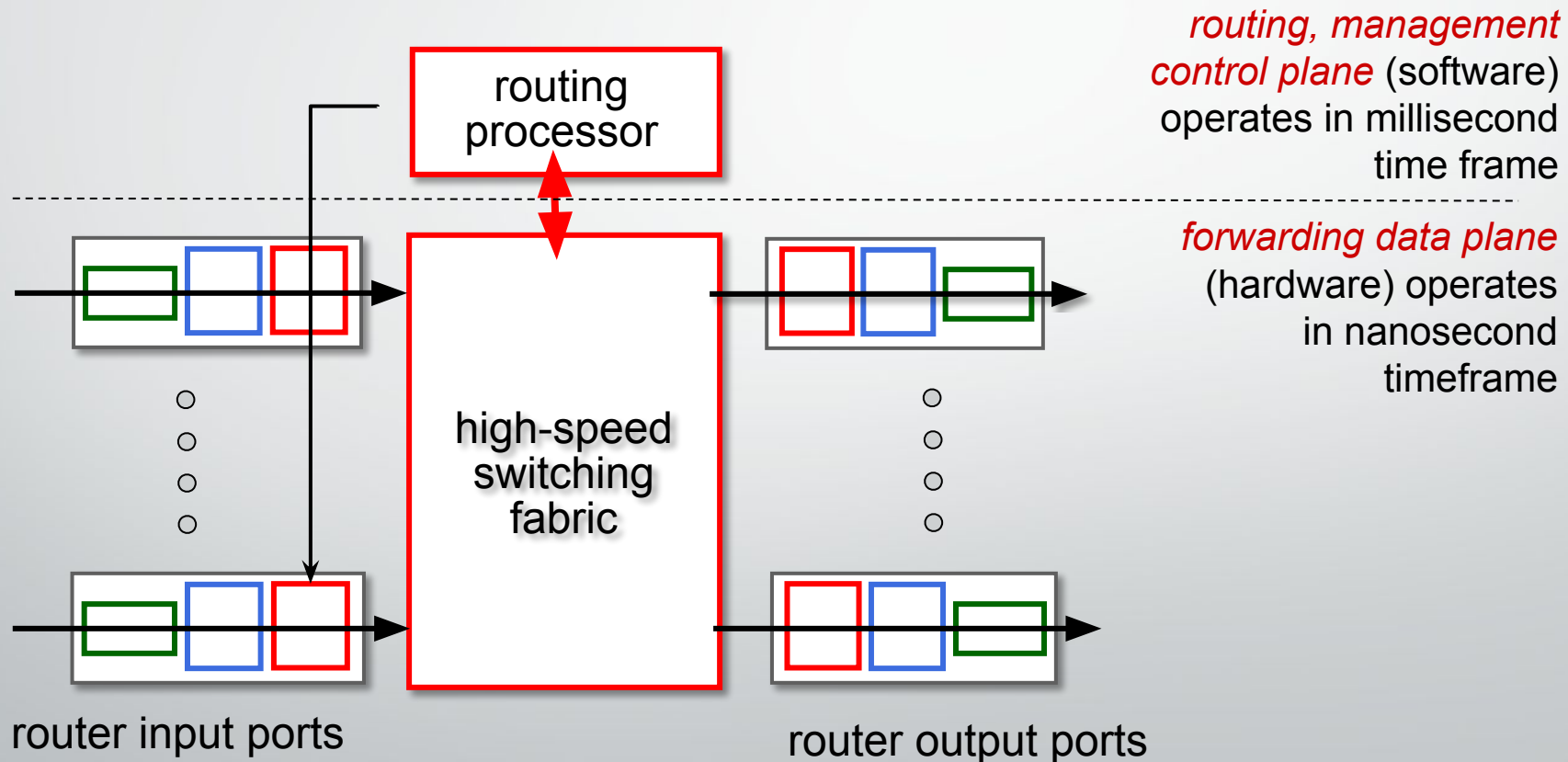
- **Packet Forwarding:**

All packets follow the same path through the network.

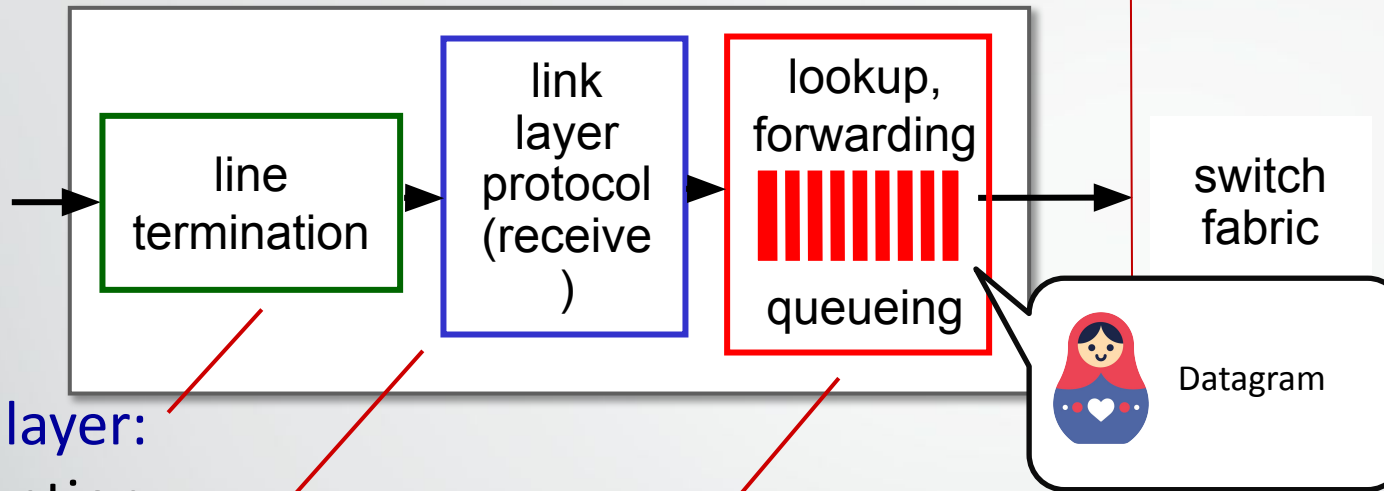


Router architecture overview

high-level view of generic router architecture:



Input port functions



physical layer:
bit-level reception

link layer:
e.g., Ethernet
(chapter 6)

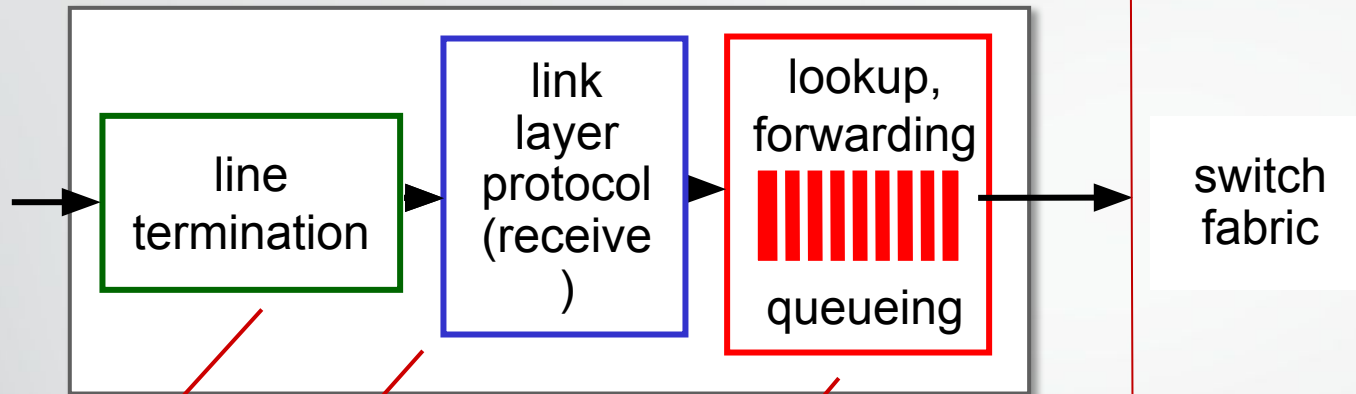


Frame

decentralized switching:

- using header field values, lookup output port using forwarding table in input port memory (*"match plus action"*)
- goal: complete input port processing at 'line speed'
- **input port queueing:** if datagrams arrive faster than forwarding rate into switch fabric

Input port functions



physical layer:
bit-level reception

link layer:
e.g., Ethernet
(chapter 6)

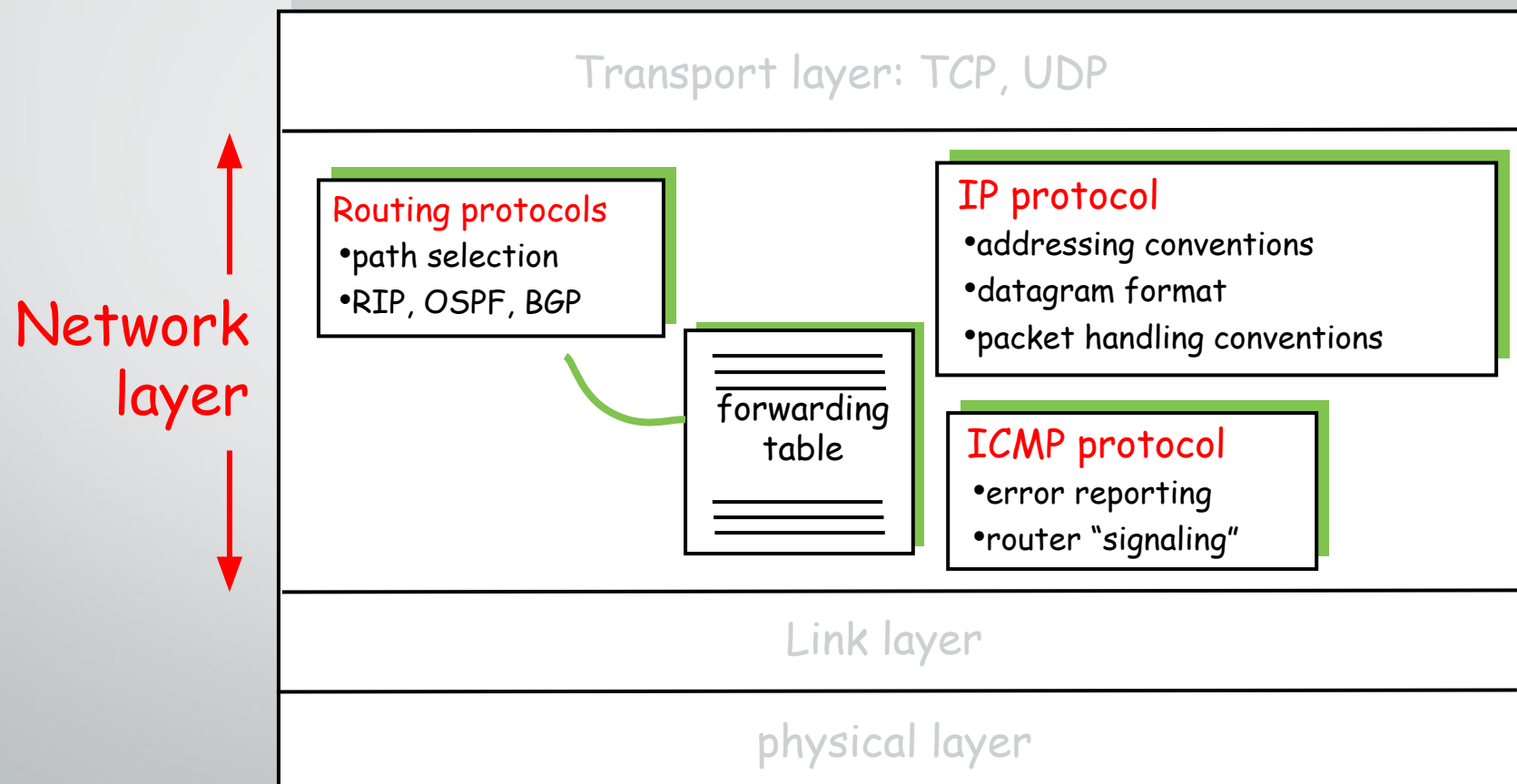
decentralized switching:

- using header field values, lookup output port using forwarding table in input port memory (*"match plus action"*)
- **destination-based forwarding:** forward based only on destination IP address (traditional)
- **generalized forwarding:** forward based on any set of header field values

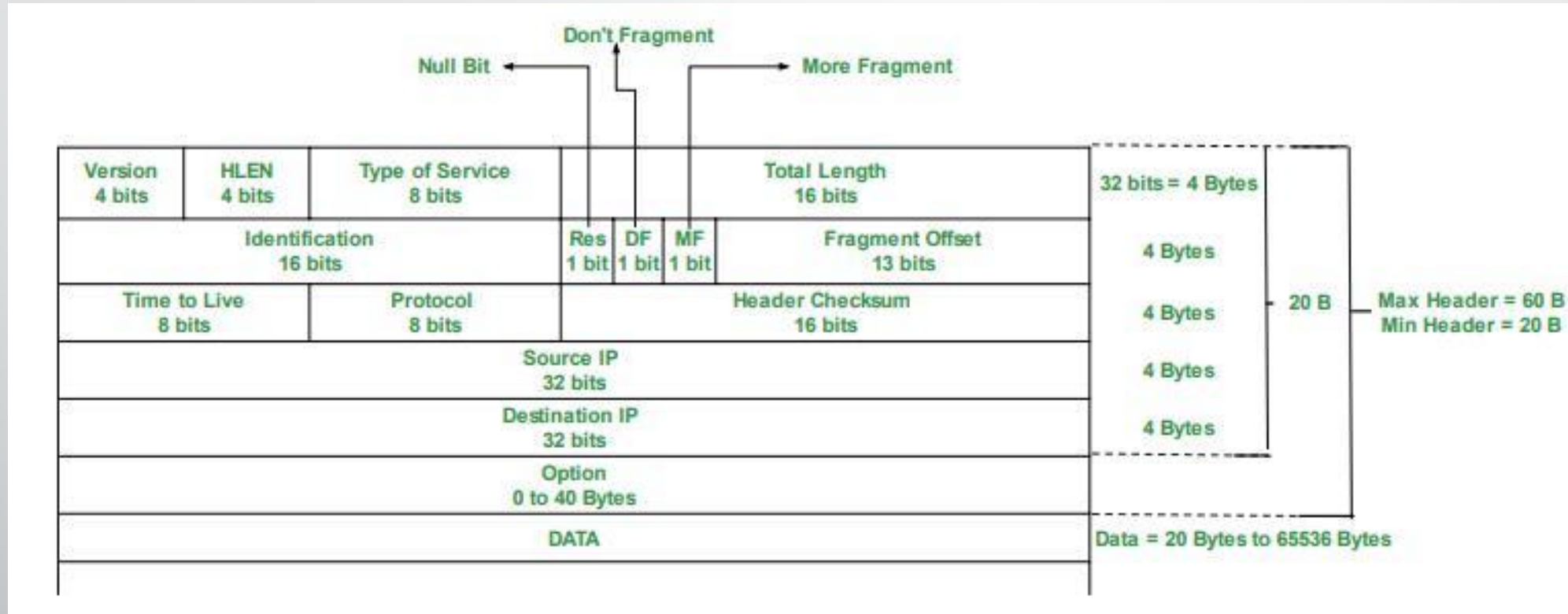
Internet Protocol IPv4

Internet Network Layer

- Host, router network layer functions:



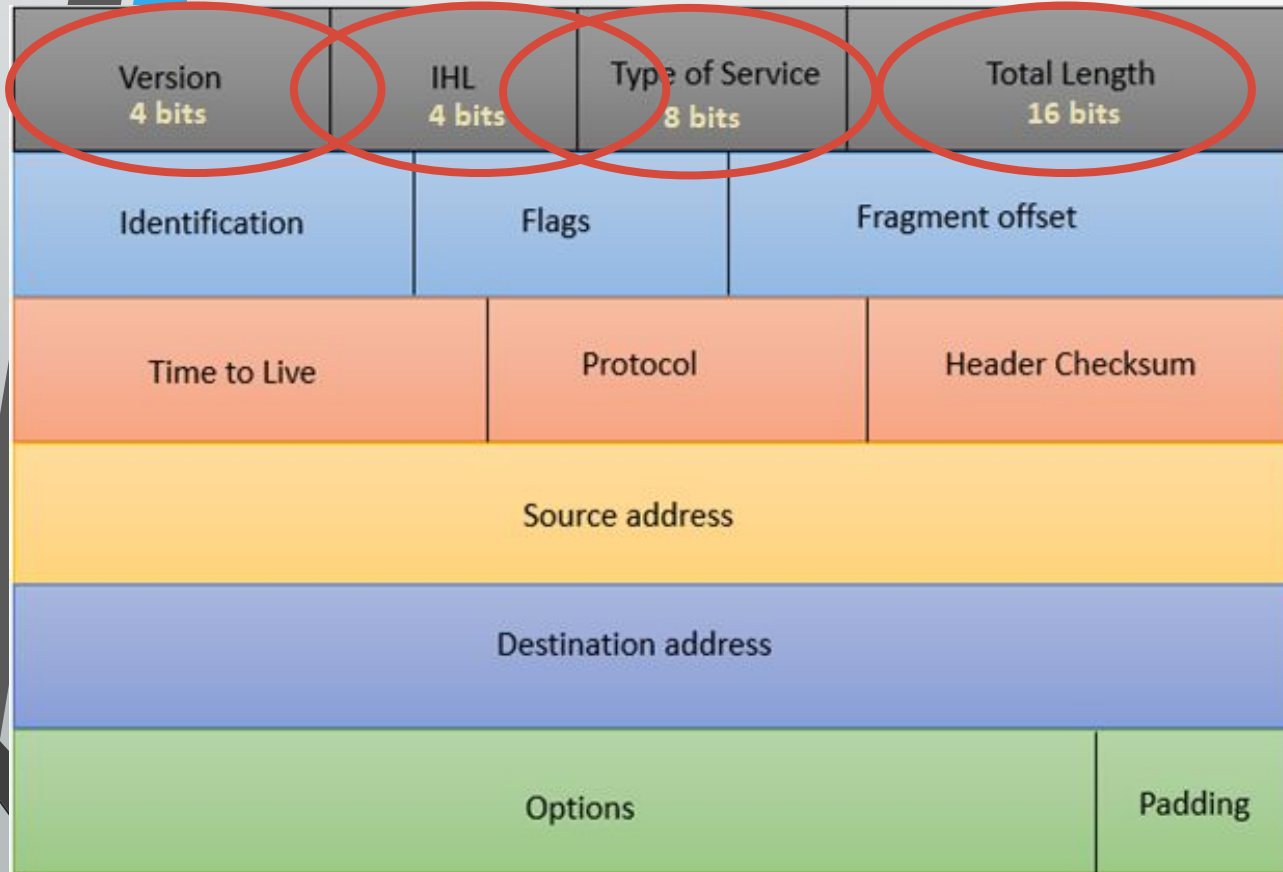
IPv4 Datagram Format



The size of an IP datagram:

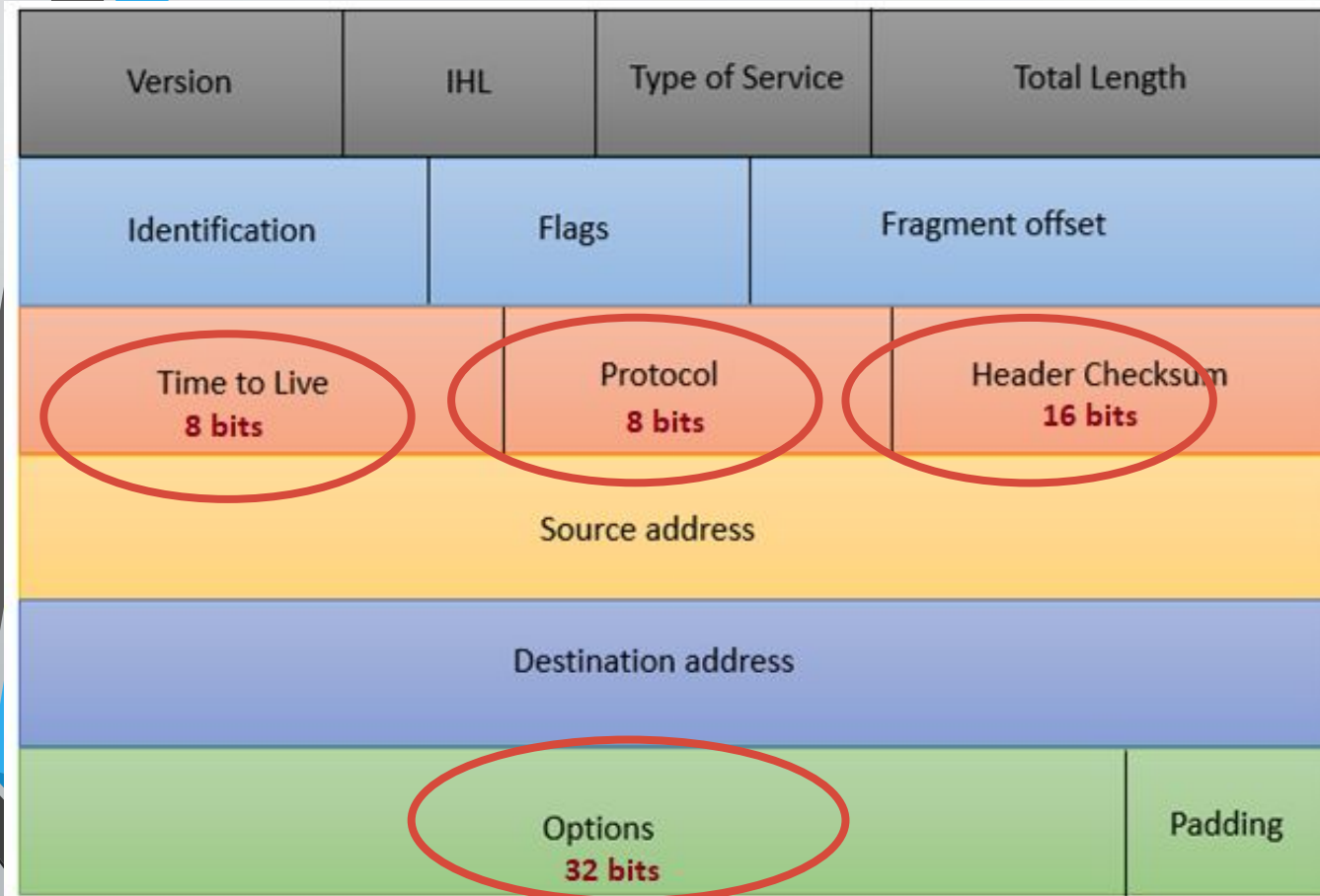
- The **minimum** size is **20 bytes** (if you have no data)
- The **maximum** size is **65,535 bytes**

IPv4 Datagram Header Format



- **Version:** value of which IP version is being used. For IPv4 the value will be 4 here.
- **Internet Header Length:** value of the header length, min 20 bytes, max 60 bytes. Shown in 4 byte word. **So min value 5, max 15.**
- **Type of Service:** for QoS (Quality of Service). To mark the packet to give special treatment or priority.
- **Total Length:** value of the entire size of the IP packet (header and data) in bytes.

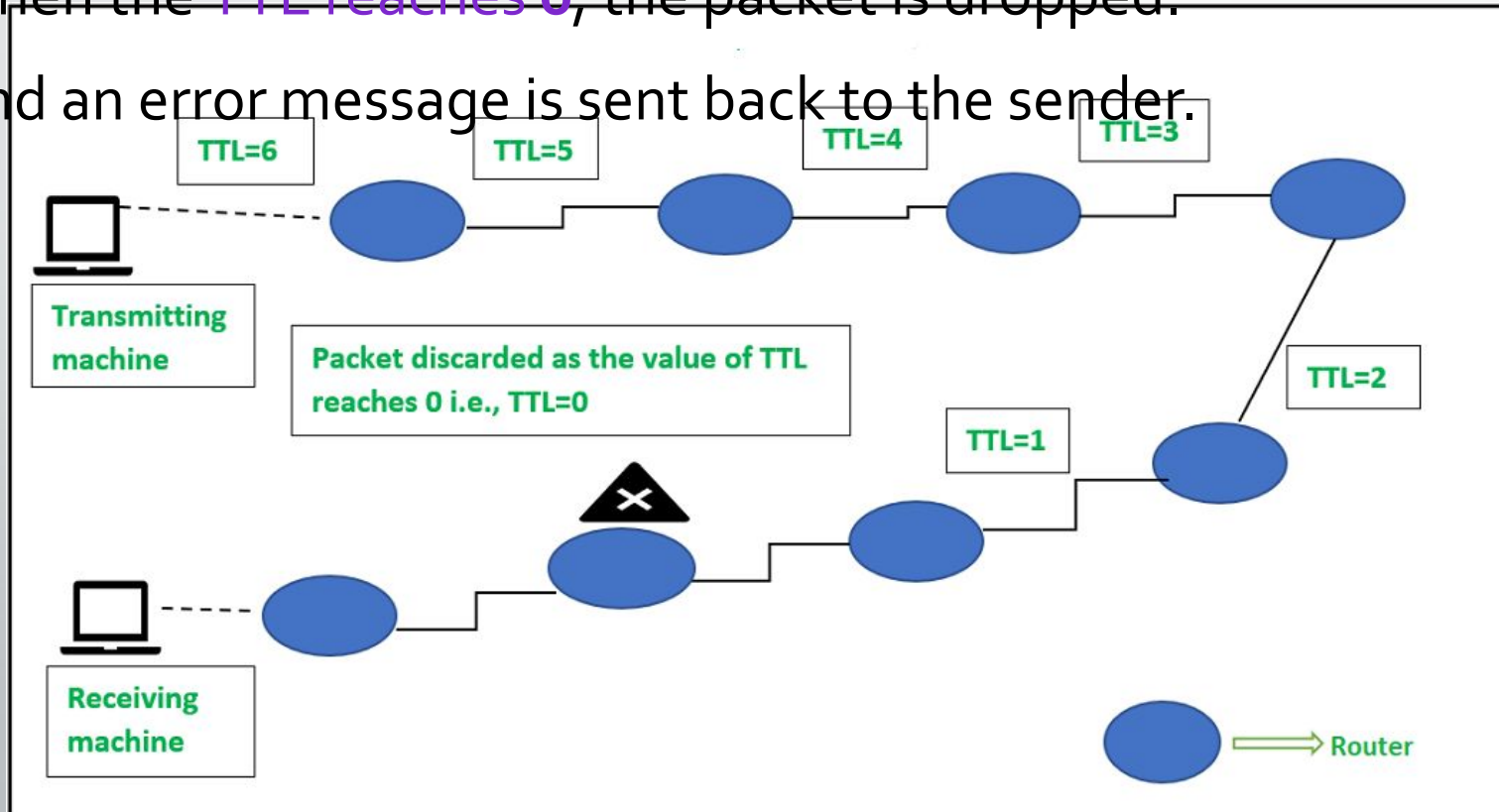
IPv4 Datagram Header Format



- **Time to Live:** maximum number of **hops** (routers) a packet can travel
- **Protocol:** value tells us which upper layer protocol is present, for example **TCP** has value **6** and **UDP** has value **17**.
- **Header Checksum:** to check if there are any errors in the header.
- **Options:** value of any extra information

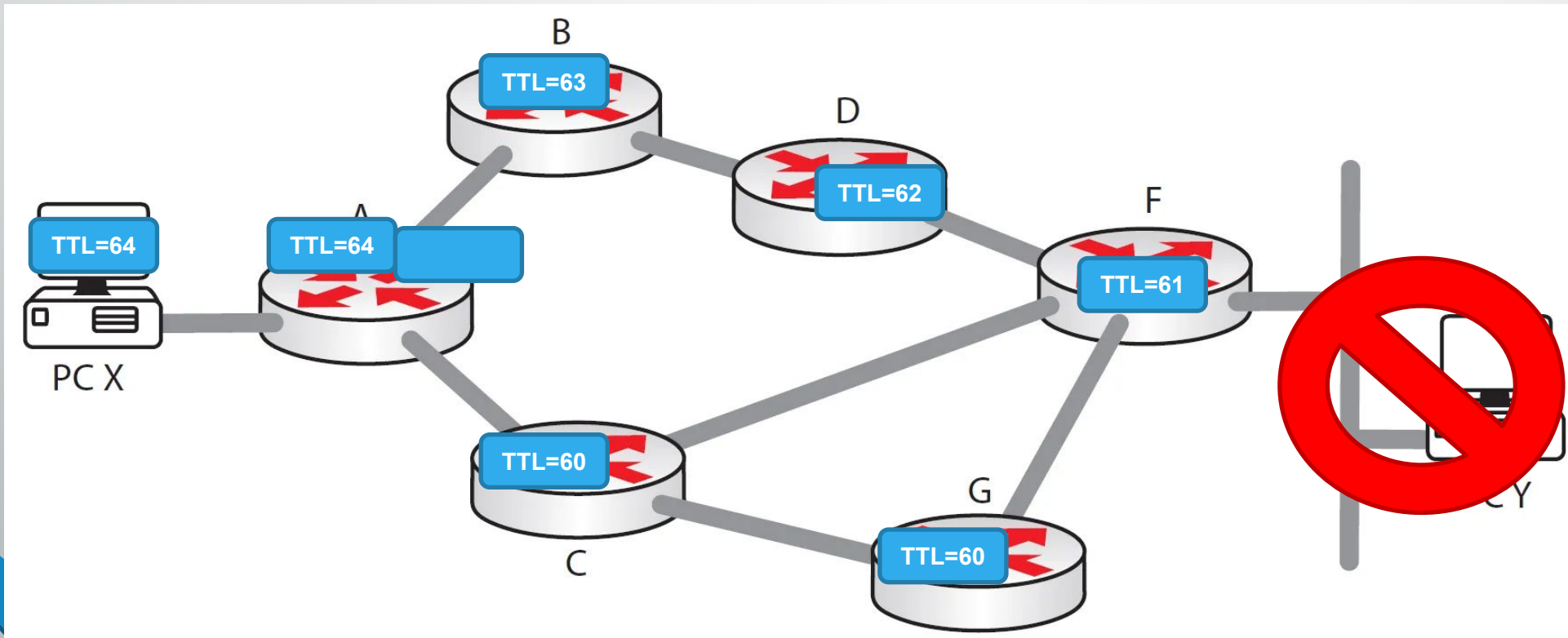
Time to Live - TTL

- Maximum number of **hops** (routers) a packet can traverse before being discarded.
- At each hop, the TTL is decreased by **1**.
- When the **TTL reaches 0**, the packet is dropped.
- And an error message is sent back to the sender.

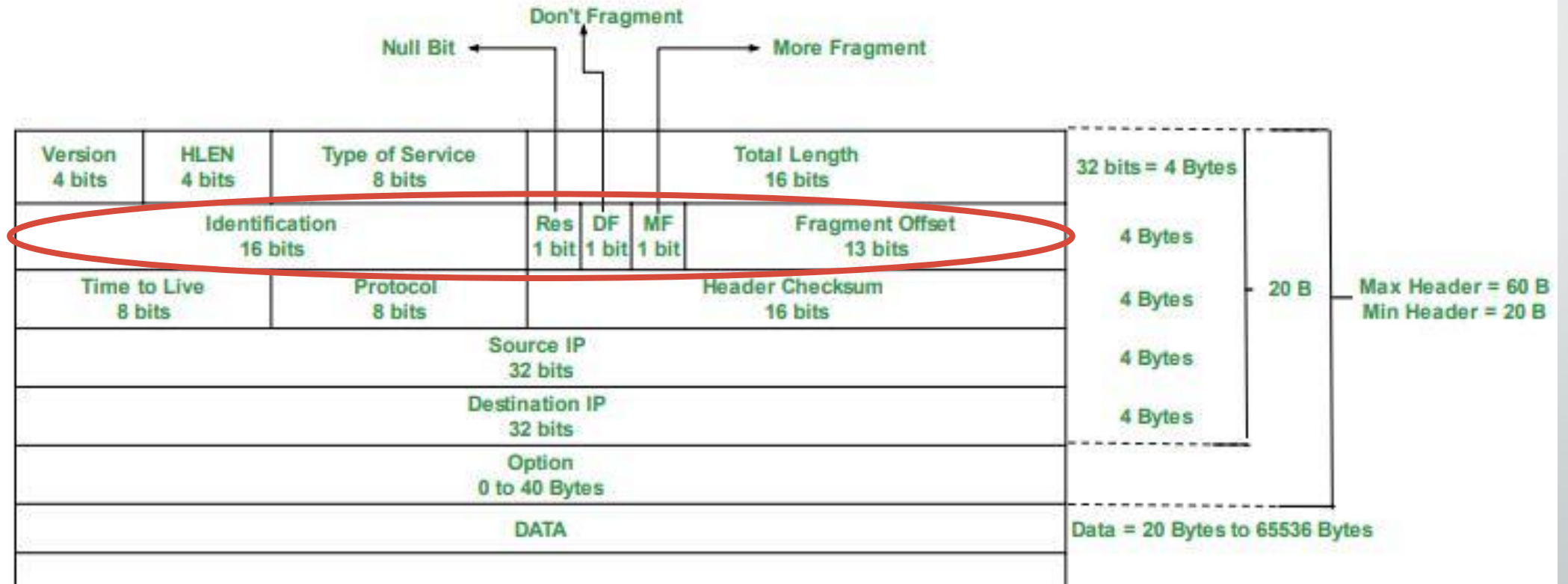


Time to Live - TTL

- **Not** just the "value of hops"
- It's a mechanism to prevent packets from **looping endlessly** in the network.
- Ensure finite packet lifetimes.

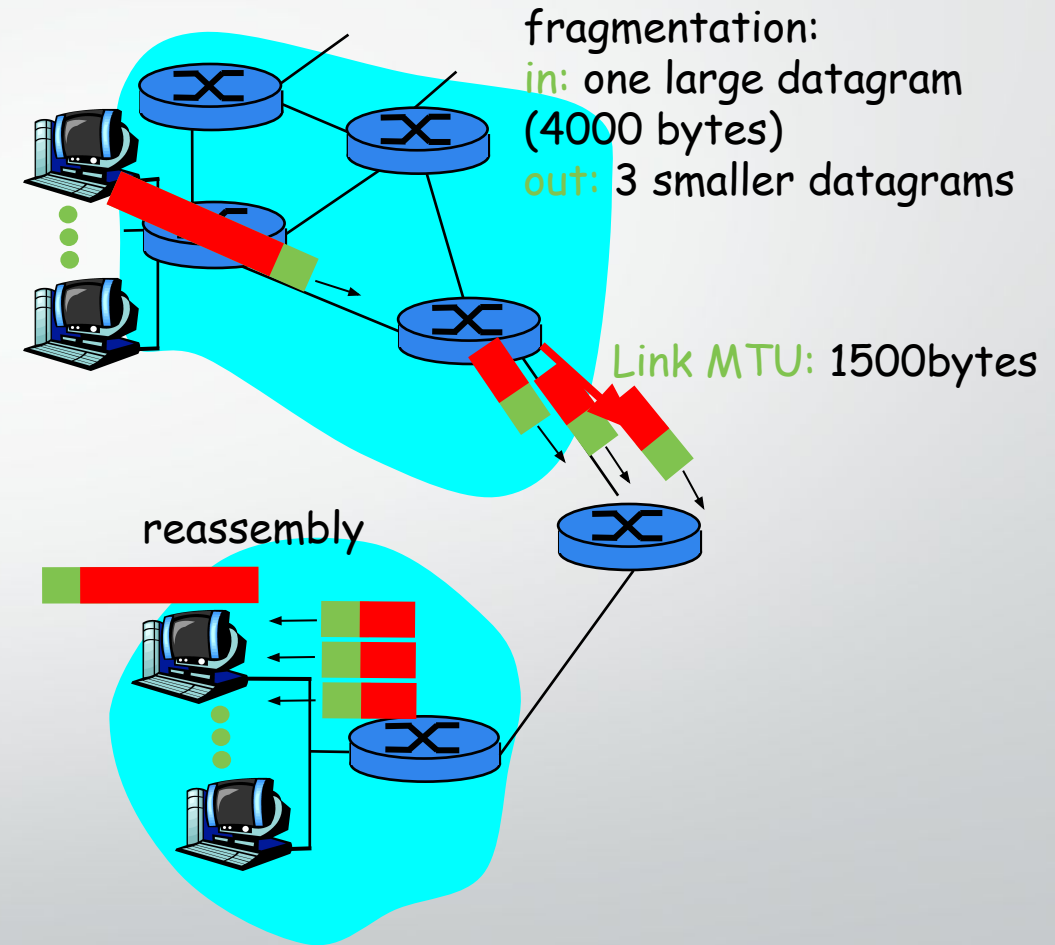


IPv4 Datagram Format



IP Fragmentation & Reassembly

- Network links have **MTU**
 - Maximum transmission unit or maximum transfer size
 - Different link types have different MTUs



IP Fragmentation & Reassembly

Original IP Datagram

Identifier	Total Length	DF May / Don't	MF Last / More	Fragment Offset
345	5140	0	0	0

IP Fragments (Ethernet)

Identifier	Total Length	DF May / Don't	MF Last / More	Fragment Offset
345	1500	0	1	0
345	1500	0	1	185
345	1500	0	1	370
345	700	0	0	555

$$MTU = 20(H) + 1480(D)$$

$$5140 = 20(H) + 5120(D)$$

$$5120 - 1480 = 3640 \text{ (1st)}$$

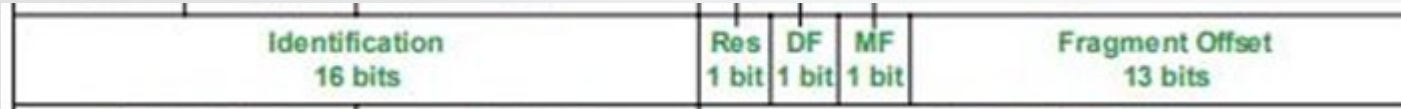
$$3640 - 1480 = 2160 \text{ (2nd)}$$

$$2160 - 1480 = 680 \text{ (3rd)}$$

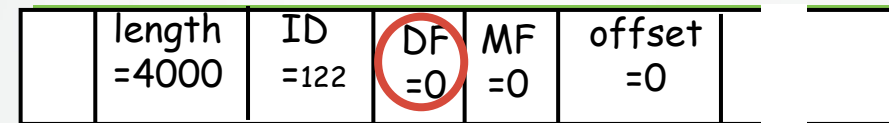
$$680 + 20 = 700$$

Data Bytes	Fragment Offset
0 - 1479	$0/8 = 0$
1480 - 2959	$1480/8 = 185$
2960 - 4439	$2960/8 = 370$
4440 - 5119	$4440/8 = 555$

IP Fragmentation & Reassembly



- Example:
 - 4000 Bytes of datagram
 - MTU = 1500 Bytes
- **DF – Don't Fragment Bit**
 - Value 0 or 1
- **Fragment Offset**
 - The value of the offset is measured in units of 8 bytes.

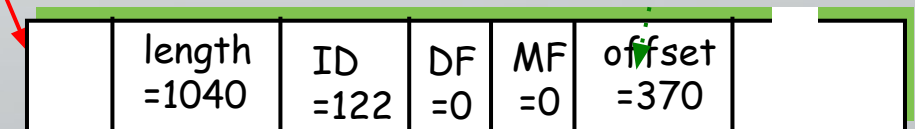
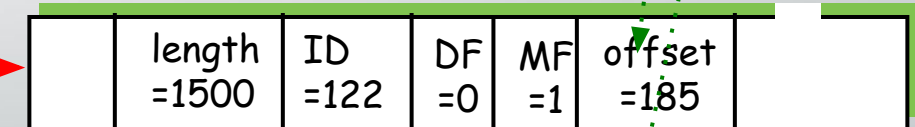
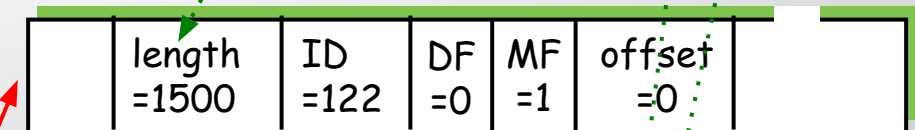


One large datagram becomes several smaller datagrams

1480 bytes in data field

offset = 1480/8

offset = 2960/8



ICMP

ICMP

- **Internet Control Message Protocol**

- Helps manage and troubleshoot IP networks.

- **Functions:**

- Reports errors in communication between devices.
 - Checks if a remote host is reachable.
 - Monitors network congestion.

- **Key Point:**

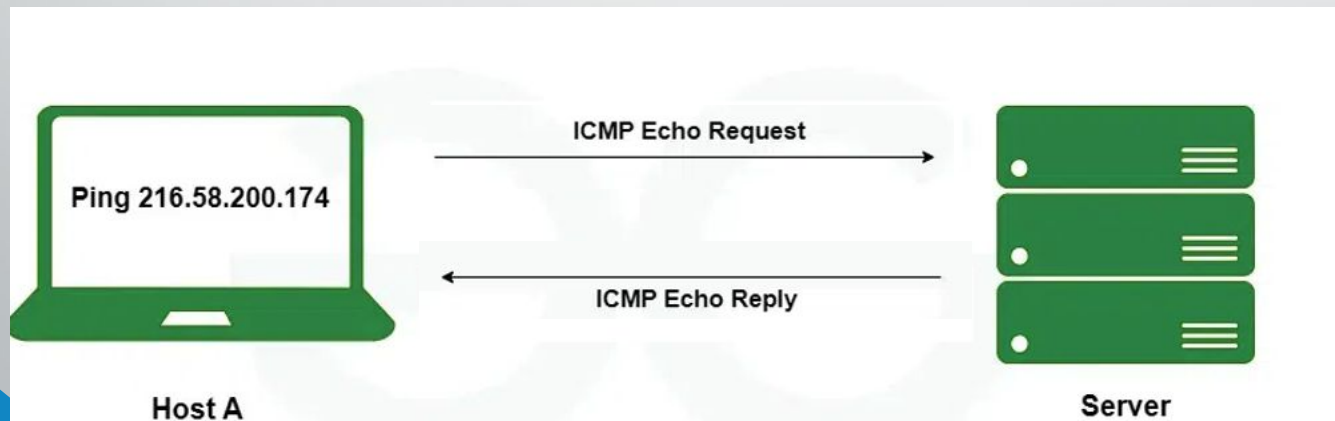
- ICMP doesn't send actual user data—it only provides network status updates.
 - Mainly used by the **operating systems** in IP network management and administration.

- **Example of ICMP in practice**

- **Ping**

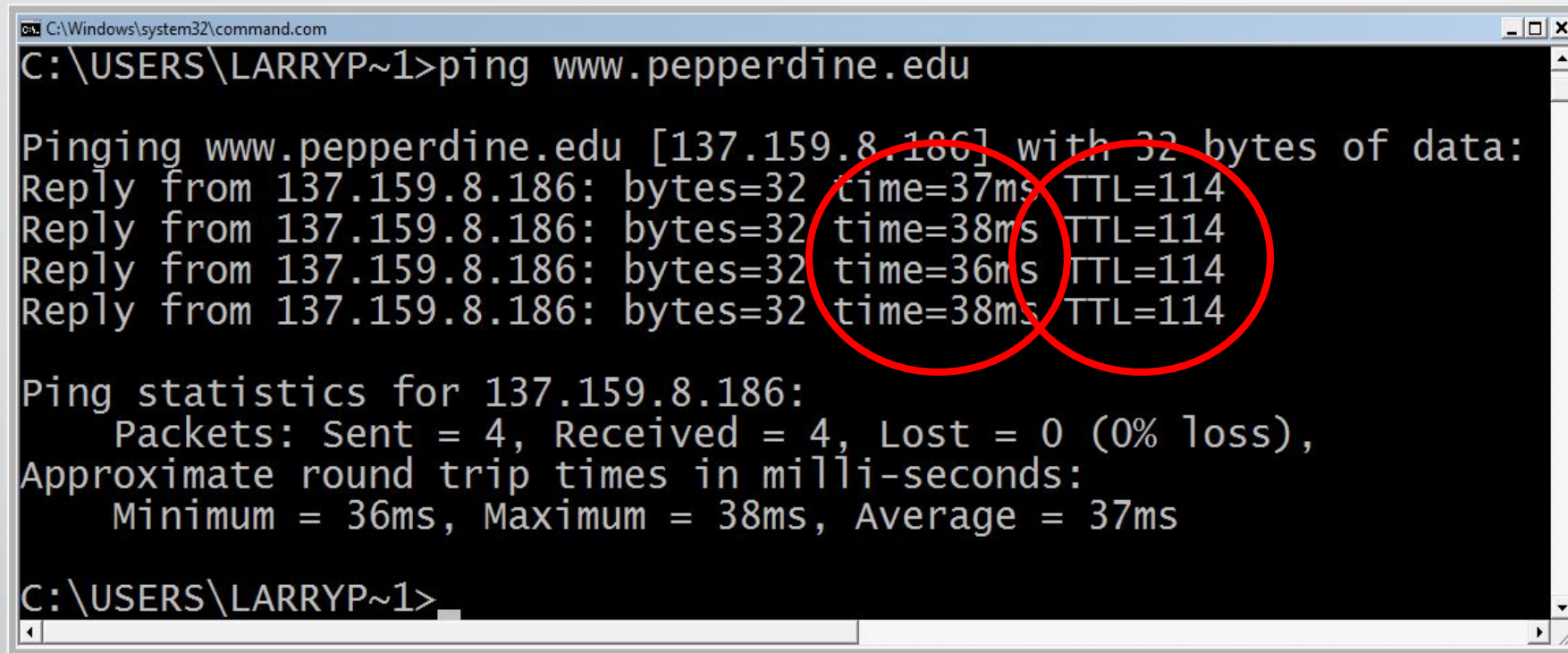
Ping

- **PING** stands for Packet Internet Groper and is a network utility tool.
 - **Purpose:** Checks if a device is reachable and measures the time it takes for data to travel (round-trip time).
 - **Mechanism:** Sends ICMP Echo Request packets to a target and waits for Echo Reply packets.
 - **Results:** Displays the number of packets sent, received, lost, and the time taken for the round-trip.



- Commands:
ping 216.58.200.174

Ping



```
C:\Windows\system32\command.com
C:\USERS\LARRYP~1>ping www.pepperdine.edu

Pinging www.pepperdine.edu [137.159.8.186] with 32 bytes of data:
Reply from 137.159.8.186: bytes=32 time=37ms TTL=114
Reply from 137.159.8.186: bytes=32 time=38ms TTL=114
Reply from 137.159.8.186: bytes=32 time=36ms TTL=114
Reply from 137.159.8.186: bytes=32 time=38ms TTL=114

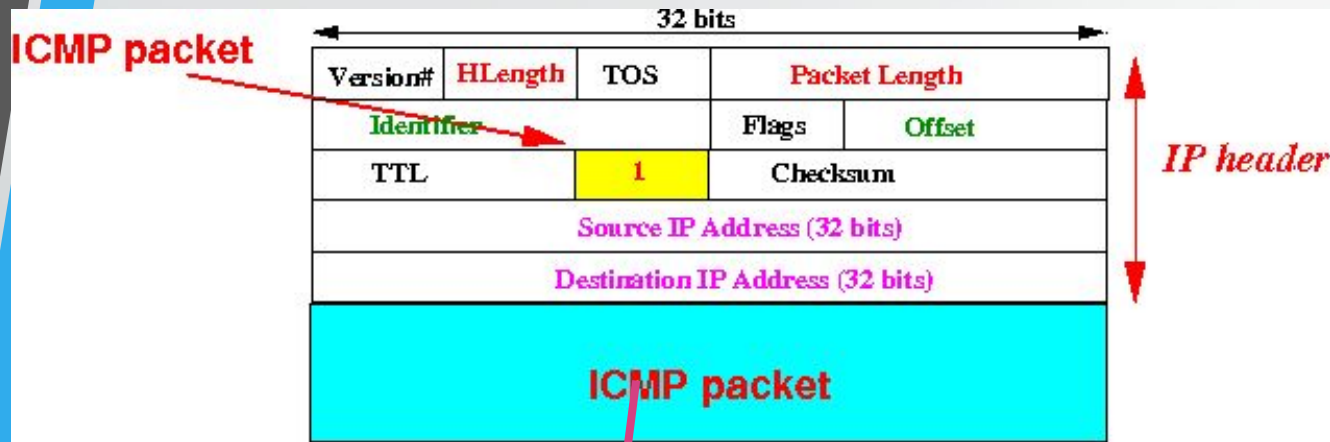
Ping statistics for 137.159.8.186:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 36ms, Maximum = 38ms, Average = 37ms

C:\USERS\LARRYP~1>
```

● Questions :

- Why 4 replies?
- What the time refer to?

ICMP Packet Format



ICMP Message Format

Type	Code	Description
0 – Echo Reply	0	Echo reply
3 – Destination Unreachable	0	Destination network unreachable
	1	Destination host unreachable
	2	Destination protocol unreachable
	3	Destination port unreachable
	4	Fragmentation needed and DF flag set
	5	Source route failed
5 – Redirect Message	0	Redirect datagram for the Network
	1	Redirect datagram for the host
	2	Redirect datagram for the Type of Service and Network
	3	Redirect datagram for the Service and Host
8 – Echo Request	0	Echo request
9 – Router Advertisement	0	Use to discover the addresses of operational routers
10 – Router Solicitation	0	
11 – Time Exceeded	0	Time to live exceeded in transit
	1	Fragment reassembly time exceeded
12 – Parameter Problem	0	Pointer indicates error
	1	Missing required option
	2	Bad length
13 – Timestamp	0	Used for time synchronization
14 – Timestamp Reply	0	Reply to Timestamp message

Unsuccessful Ping

```
C:\>ping 10.2.104.2
```

```
Pinging 10.2.104.2 with 32 bytes of data:
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

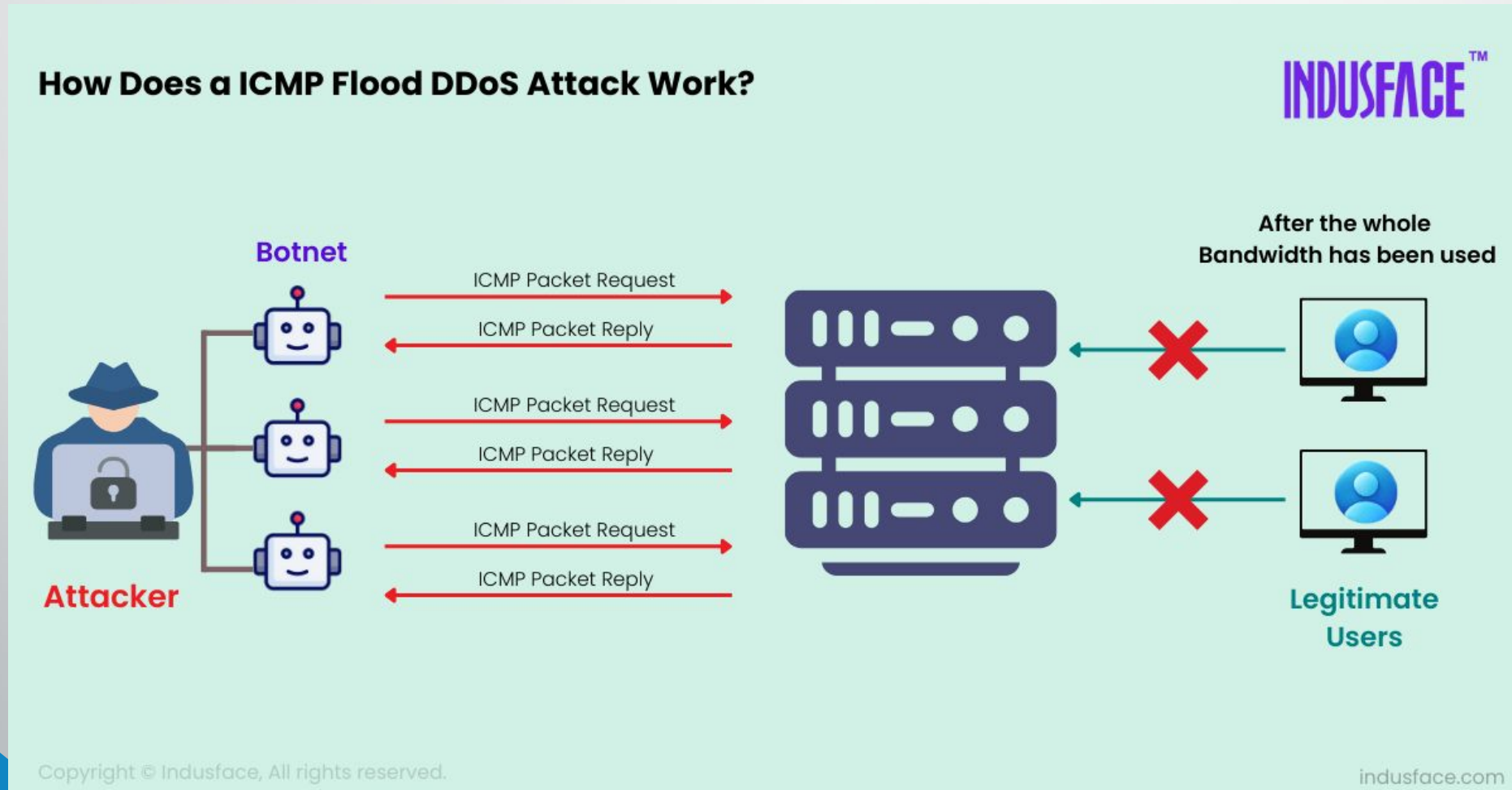
```
Request timed out.
```

```
Ping statistics for 10.2.104.2:
```

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

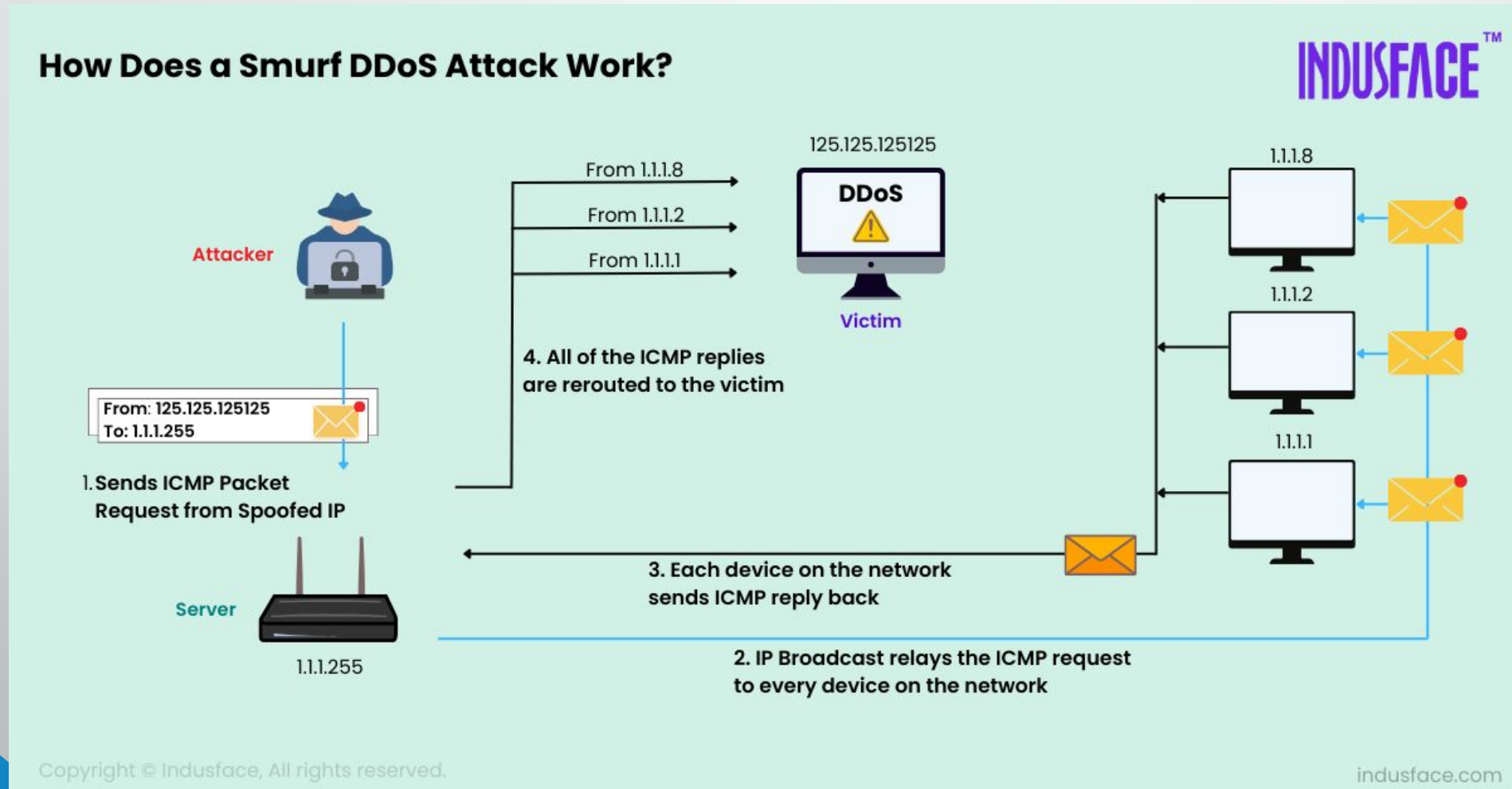
Ping Attacks

- ICMP DDOS attack – Zombie Attack:

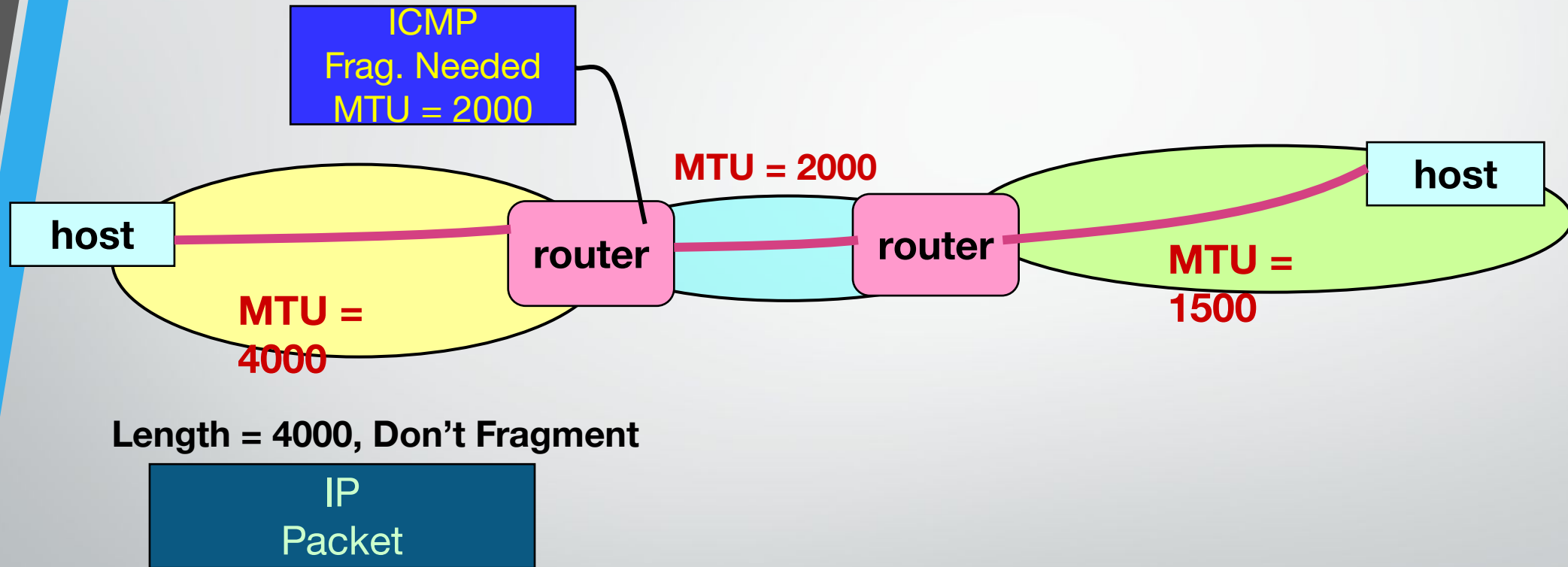


Ping Attacks

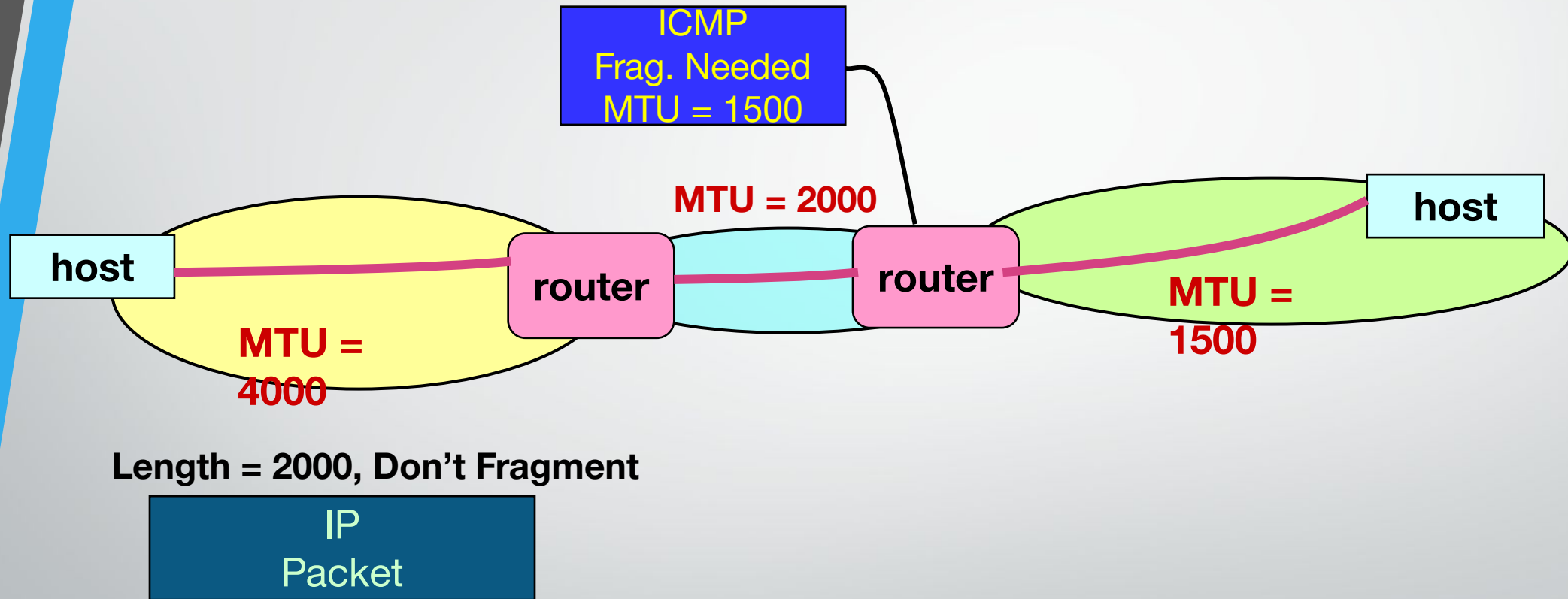
- ICMP DDOS attack – Packet magnification (or ICMP Smurf):



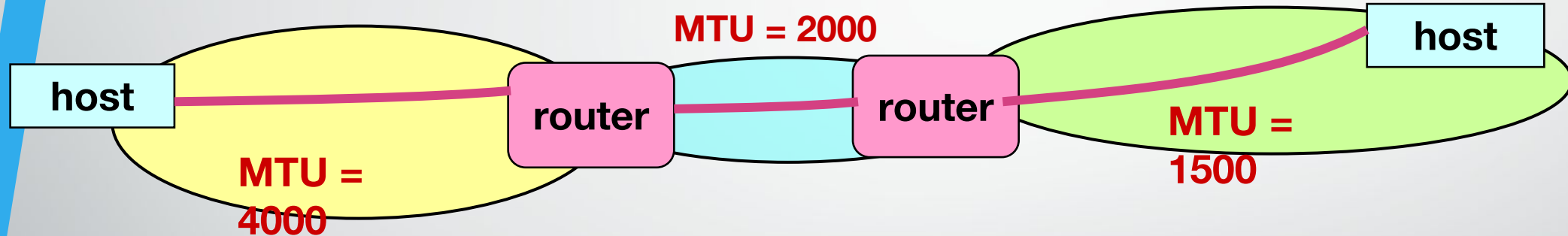
IP MTU Discovery with ICMP



IP MTU Discovery with ICMP



IP MTU Discovery with ICMP



Length = 1500, Don't Fragment

IP
Packet

- When successful, no reply at IP level
“No news is good news”
Higher level protocol might have some form of acknowledgement

Traceroute

- A network diagnostic tool used to trace the path that data packets take from your computer to a target server or IP address.
 - **Purpose:** Identifies the routers or hops data passes through to reach its destination mostly for troubleshooting
 - **Mechanism:** Uses ICMP or UDP packets with incrementing TTL (Time-to-Live) values to get "Time Exceeded" responses from each hop.
 - **Results:** Displays the IP address, hostname (if resolvable), and latency for each hop.
- Commands:
 - Unix: **traceroute**
 - Cisco IOS: **traceroute (trace)**
 - DOS: **tracert**



```

C:\Command Prompt
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\Users\skazi>tracert www.yahoo.com
'tracert' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\skazi>tracert www.yahoo.com

Tracing route to me-ycpi-cf-www.g06.yahoodns.net [27.123.42.205]
over a maximum of 30 hops:

  1    4 ms    1 ms    1 ms    172.18.192.1
  2    *      *      *      Request timed out.
  3    3 ms    1 ms    1 ms    172.31.2.129
  4    1 ms    1 ms    1 ms    10.151.6.89
  5    1 ms    2 ms    1 ms    10.0.100.5
  6    2 ms    1 ms    1 ms    202.4.100.253
  7    1 ms    2 ms    2 ms    GI0-2-2-aggr01.as58656.net [103.12.177.1]
  8    2 ms    2 ms    2 ms    10.12.176.237
  9    3 ms    2 ms    2 ms    103.16.155.149
 10    2 ms    1 ms    1 ms    103.16.152.30
 11   11 ms   11 ms   10 ms   103.16.152.82
 12    *      51 ms   51 ms   103.16.153.21
 13   51 ms   51 ms   51 ms   103.16.153.18
 14   57 ms   57 ms   57 ms   ae6-1538.rt.eqx.sin.sg.retn.net [87.245.240.208]
 15   62 ms   63 ms   62 ms   ix-be-20.ecore4.esin4-singapore.as6453.net [180.87.54.66]
 16   64 ms   65 ms   64 ms   if-bundle-18-2.qcore2.esin4-singapore.as6453.net [180.87.108.80]
 17   70 ms   70 ms   71 ms   180.87.55.59
 18    *      *      *      Request timed out.
 19   69 ms   70 ms   78 ms   14.143.59.46.static-mumbai.vsnl.net.in [14.143.59.46]
 20   68 ms   68 ms   67 ms   e2-ha.ycpi.ina.yahoo.com [27.123.42.205]

Trace complete.

C:\Users\skazi>_

```

Using Tracert

Hop 1: Our local router or gateway (private IP address).

Hops 2–5: Internal routing within Bracu ISP's private network (non-public IPs).

Hop 6: First public IP, ISP's gateway to the internet.

Hops 7–9: Routing through regional and backbone ISPs.

Hops 10–13: Routing through Singapore (a major internet hub).

Hops 14–19: Routing through Indian networks, ending in Mumbai.

Hop 20: Final destination—Yahoo's server, located in India, near Mumbai.

Traceroute: Another example

Hop 1: User LAN router

Hops 2-4: Verizon network (a backbone ISP)

Hops 5-6: Alternet (a backbone ISP)

Hops 7-11: Level 3 (a backbone ISP)

Hops 12-14: the Google LAN

```
C:\Windows\system32\COMMAND.com
C:\USERS\LARRYP~1>tracert www.google.com

Tracing route to www.l.google.com [74.125.19.147]
over a maximum of 30 hops:

  1    3 ms    1 ms    1 ms  192.168.1.1
  2   38 ms   37 ms   37 ms  L100.LSANCA-DSL-14.verizon-gni.net [71.105.96.11]
  3   38 ms   34 ms   36 ms  P1-3.LSANCA-LCR-03.verizon-gni.net [130.81.35.81]
  4   34 ms   37 ms   34 ms  so-6-1-2-0.LAX01-BB-RTR1.verizon-gni.net [130.81.28.225]
  5   37 ms   35 ms   38 ms  0.so-1-3-0.XL3.LAX15.ALTER.NET [152.63.114.145]
  6   36 ms   36 ms   40 ms  0.ge-6-0-0.BR2.LAX15.ALTER.NET [152.63.116.149]
  7   38 ms   40 ms   40 ms  xe-11-0-0.edge1.SanJose3.level3.net [4.68.111.249]
  8   46 ms   38 ms   49 ms  ae-73-70.ebr3.LosAngeles1.Level3.net [4.69.144.116]
  9   47 ms   55 ms   52 ms  ae-2.ebr3.SanJose1.Level3.net [4.69.132.91]
 10   68 ms   54 ms  126 ms  ae-63-63.csw1.SanJose1.Level3.net [4.69.134.226]
 11   72 ms   45 ms  115 ms  ae-1-69.edge1.SanJose1.Level3.net [4.68.18.14]
 12  137 ms   51 ms   49 ms  GOOGLE-INC.edge1.SanJose1.Level3.net [4.79.43.146]
 13   49 ms   49 ms   54 ms  209.85.251.98
 14   47 ms   47 ms   46 ms  nuq04s01-in-f147.1e100.net [74.125.19.147]

Trace complete.
```

Traceroute: Request Timed Out

This message indicates that the router security settings keep it from revealing its identity or the router and connection are slow.

```
* * * Request timed out.
```

The End